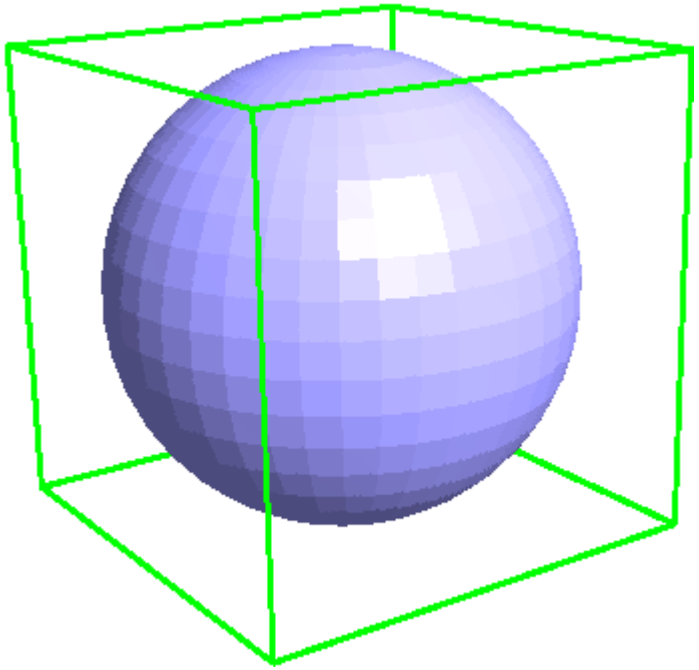
Vector Tools for Computer Graphics

CS 341: Computer Graphics

Basic Objects

- We represent objects using mainly linear primitives:
  - points
  - lines, segments
  - planes, polygons
- Two branches of Mathematics
  - Vector Analysis
  - Transformation



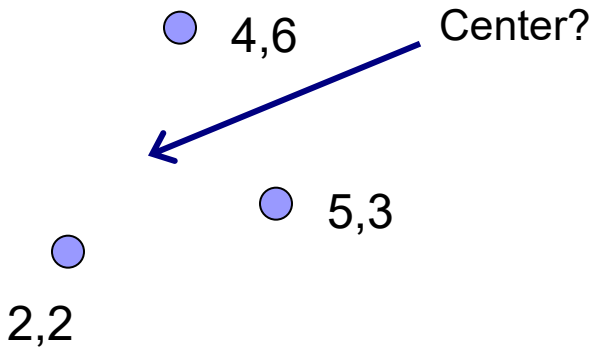
Basic Definitions

- Points specify location in space (or in the plane).
- Vectors have magnitude and direction (like velocity).

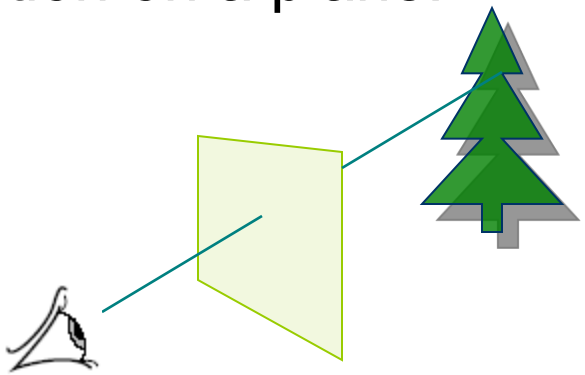
Points $\neq$ Vectors

Why Vectors?

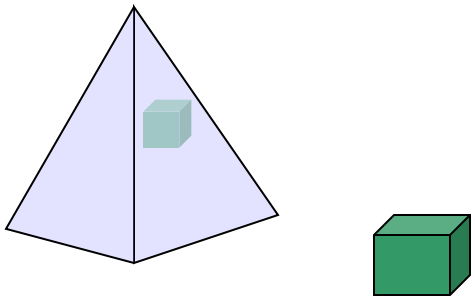
- Circle through 3 points:



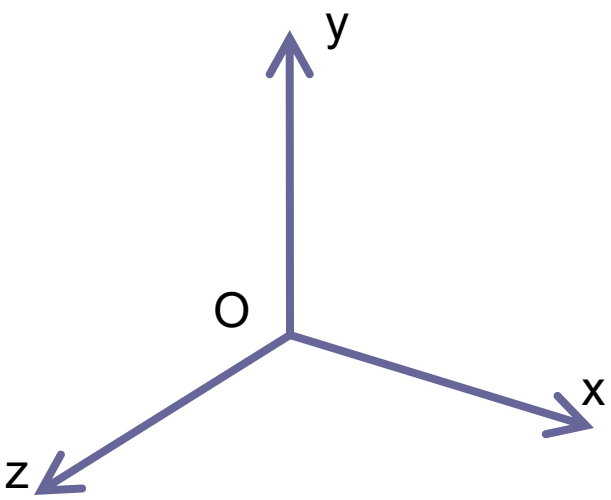
- Projection on a plane:



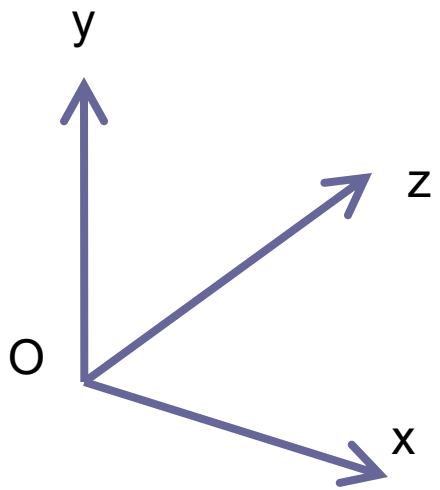
- Reflection of an object:



Coordinate Systems

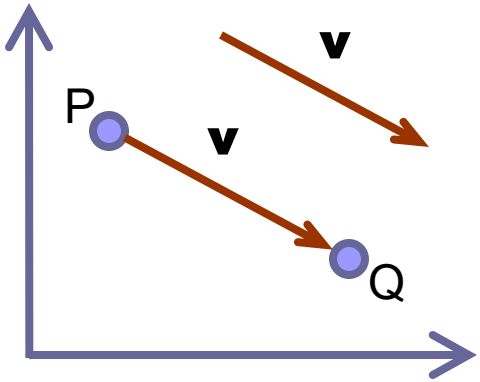


Right Handed Coordinate System



Left Handed Coordinate System

Basics of Vectors



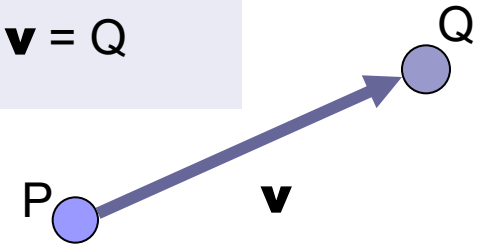
Vector as displacement:

v is a vector from point P to point Q.

The **difference** between two points is a vector: **v** = Q - P

Another way:

The **sum** of a point and a vector is a point : P + **v** = Q



Operations on Vectors

Two operations

Addition

$\mathbf{a} + \mathbf{b}$

$\mathbf{a} = (3,5,8)$, $\mathbf{b} = (-1,2,-4)$

$\mathbf{a} + \mathbf{b} = (2,7,4)$

Multiplication be scalars

$s\mathbf{a}$

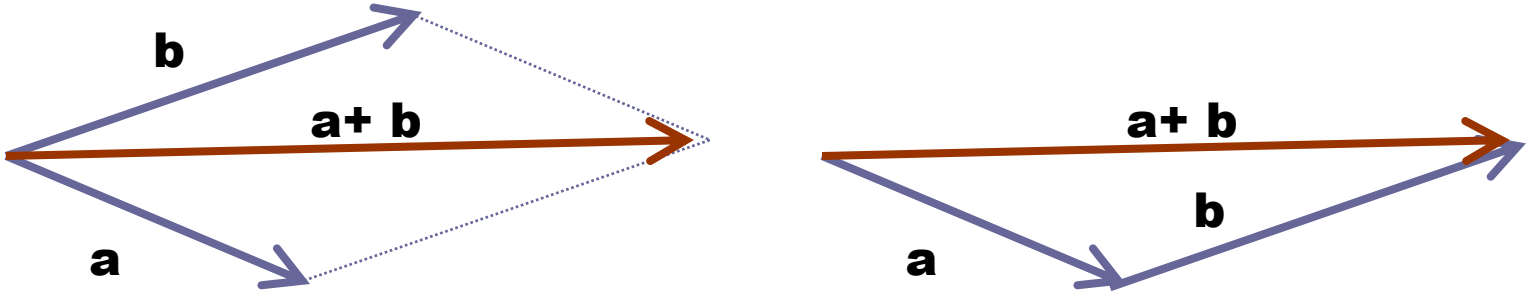
$\mathbf{a} = (3,-5,8)$, $s = 5$

$5\mathbf{a} = (15,-25,40)$

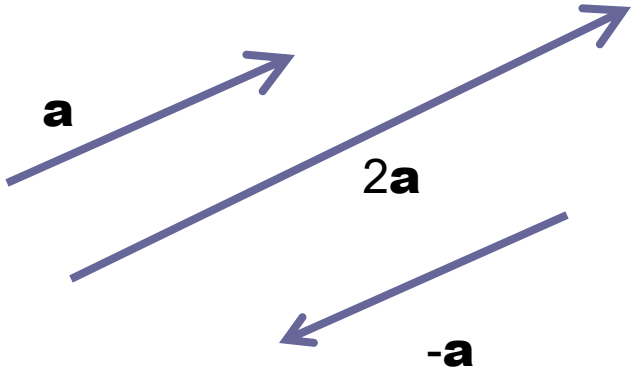
operations are done ***componentwise***

Operations on vectors

Addition

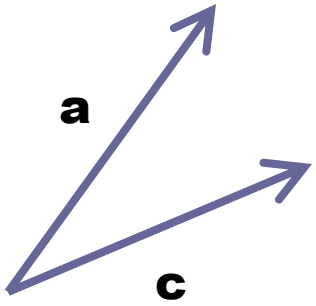


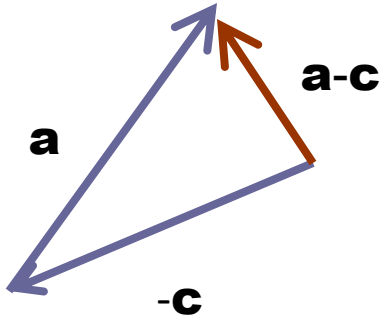
Multiplication by scalar

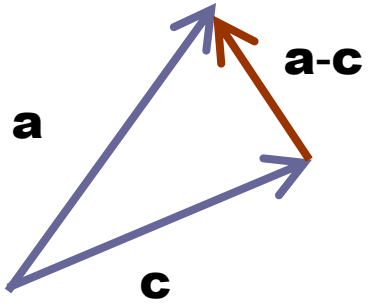


Operations on vectors

Subtraction







Linear Combination of Vectors

Definition

Linear combination of *m* vectors $\mathbf{v}_1$, $\mathbf{v}_2$, …, $\mathbf{v}_m$:

$$\mathbf{w} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \ldots + a_m\mathbf{v}_m$$

where $a_1, a_2, \ldots a_m$ are scalars.

Types

Affine Combination

$a_1 + a_2 + \ldots + a_m = 1$

Examples:

$3\mathbf{a} + 2\mathbf{b} - 4\mathbf{c}$

$(1- t)\mathbf{a} + (t)\mathbf{b}$

Convex Combination

$a_1 + a_2 + \ldots + a_m = 1$

and $a_i \geq 0$, for $i = 1, \ldots m$.

Example: $.3\mathbf{a} + .2\mathbf{b} + .5\mathbf{c}$

Properties of vectors

Length or size

$$\mathbf{w} = (w_1, w_2, \ldots, w_n)$$

$$|\mathbf{w}| = \sqrt{w_1^2 + w_2^2 + \ldots + w_n^2}$$

Unit vector

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}$$

- The process is called **normalizing**
- Used to refer **direction**

The **standard unit vectors**: **i** = (1,0,0), **j** = (0,1,0) and **k** = (0,0,1)

Dot Product

The dot product ***d*** of two vectors $\mathbf{v} = (v_1, v_2, \ldots, v_n)$ and $\mathbf{w} = (w_1, w_2, \ldots w_n)$:

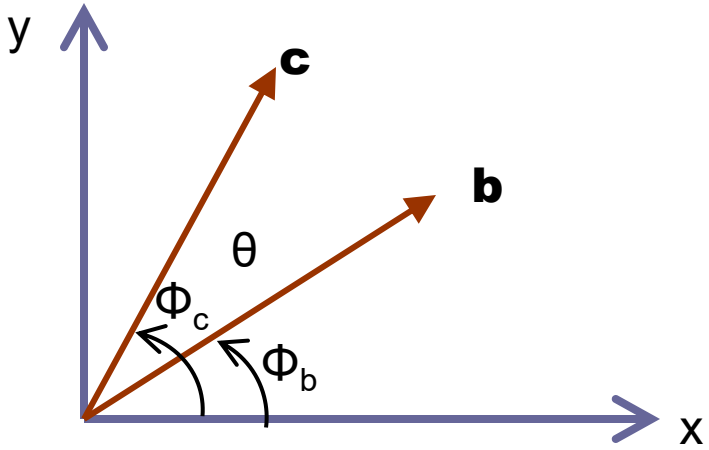
$$d = \mathbf{v}.\mathbf{w} = \sum_{i=1}^{n} v_i w_i$$

Properties

1. Symmetry: $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$
2. Linearity: $(\mathbf{a}+\mathbf{c}) \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{b} + \mathbf{c} \cdot \mathbf{b}$
3. Homogeneity: $(s\mathbf{a}) \cdot \mathbf{b} = s(\mathbf{a} \cdot \mathbf{b})$
4. $|\mathbf{b}|^2 = \mathbf{b} \cdot \mathbf{b}$

Application of Dot Product

Angle between two vectors:



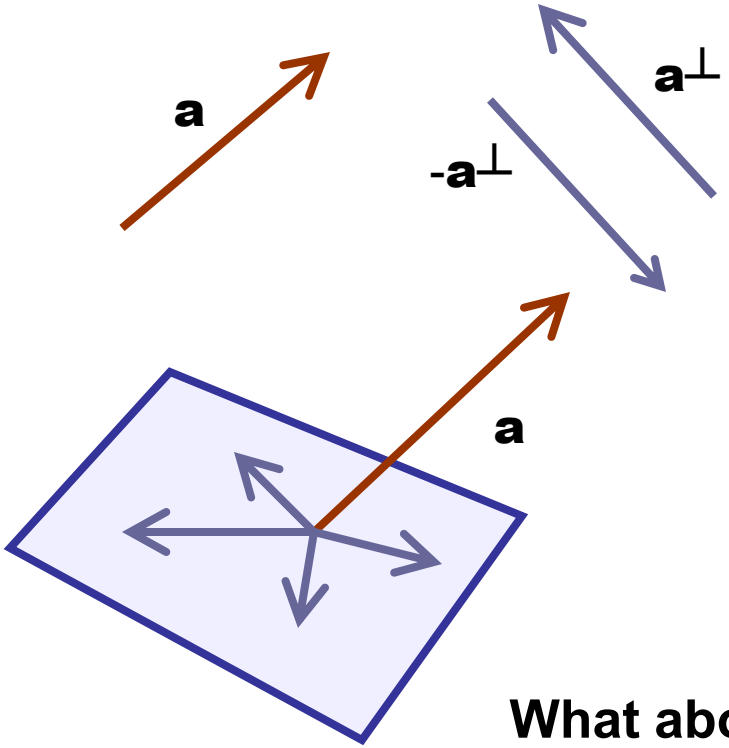
$$\cos(\theta) = \hat{\mathbf{b}} \cdot \hat{\mathbf{c}}$$

Two vectors **b** and **c** are **perpendicular** (**orthogonal/normal**) if **b**·**c** = 0

2D “perp” Vector

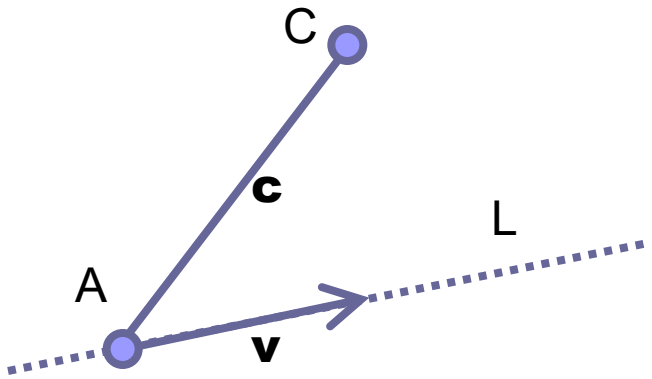
Which vector is perpendicular to the 2D vector $\mathbf{a} = (a_x, a_y)$?

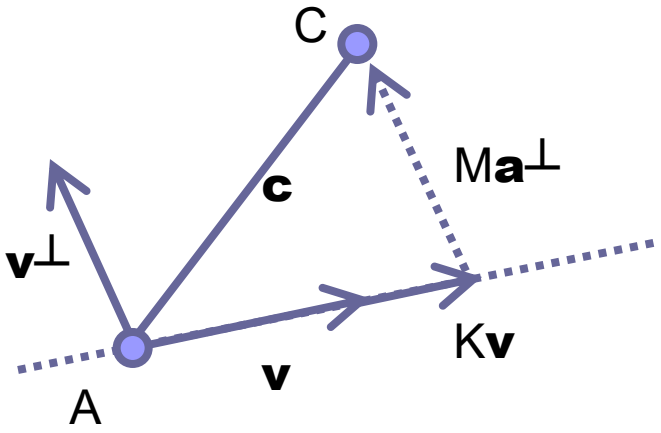
Let $\mathbf{a} = (a_x, a_y)$.
Then
$\mathbf{a}^{\perp} = (-a_y, a_x)$
is the
counterclockwise perpendicular to **a**.



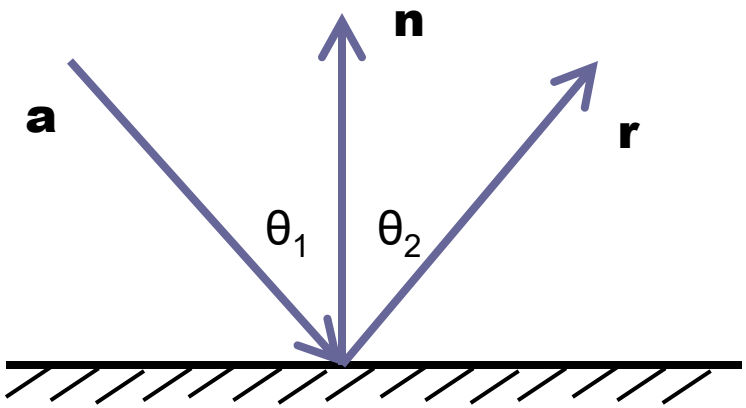
What about 3D case?

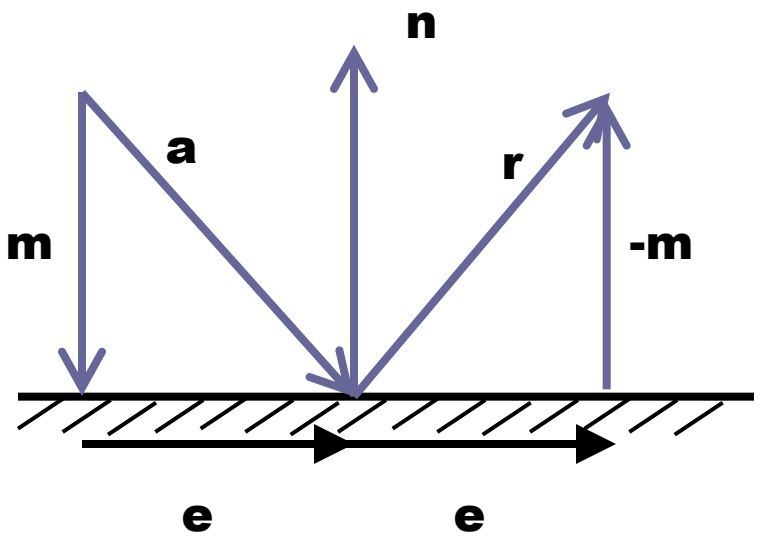
Applications: Orthogonal Projection





Applications: Reflections



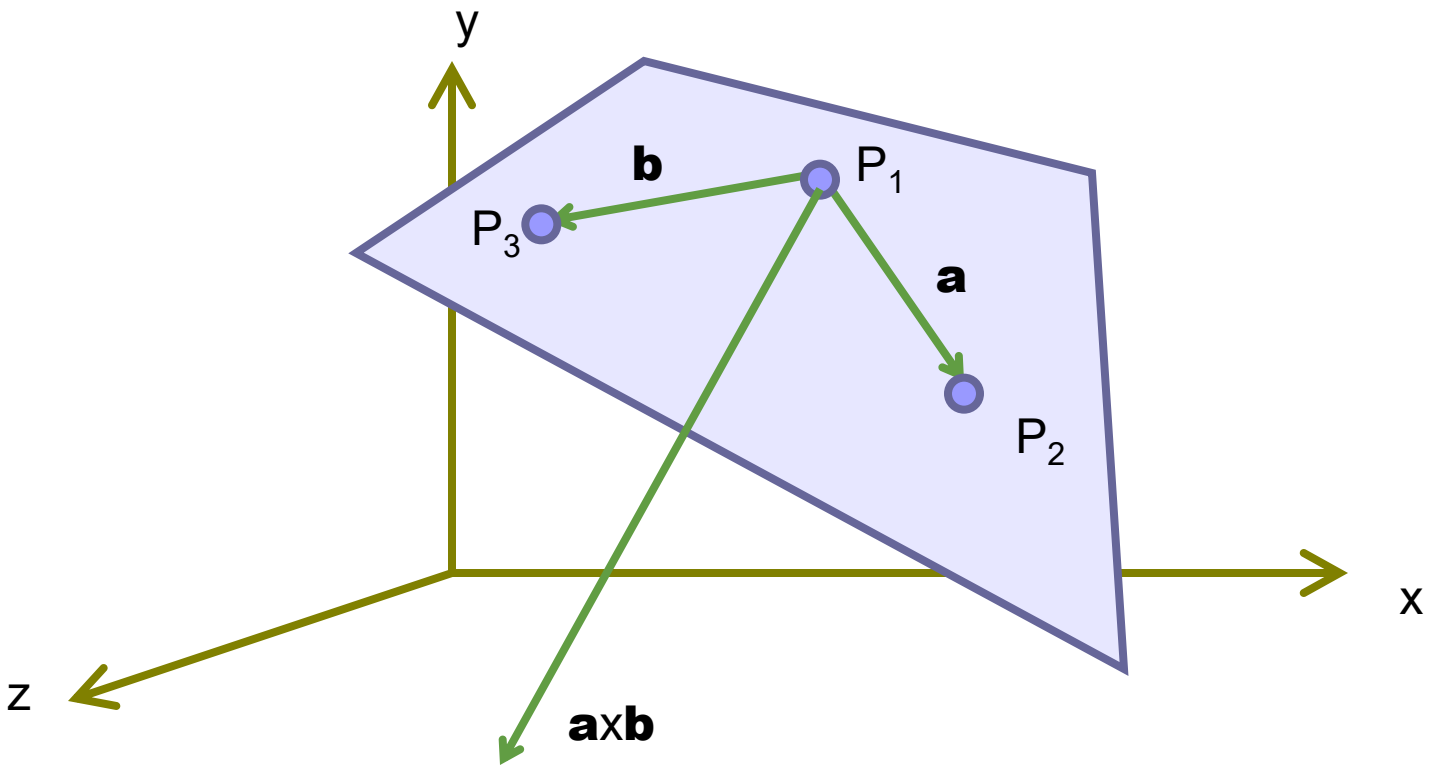


Cross Product

- Also called **vector product**.
- Defined for **3D** vectors only.

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$$

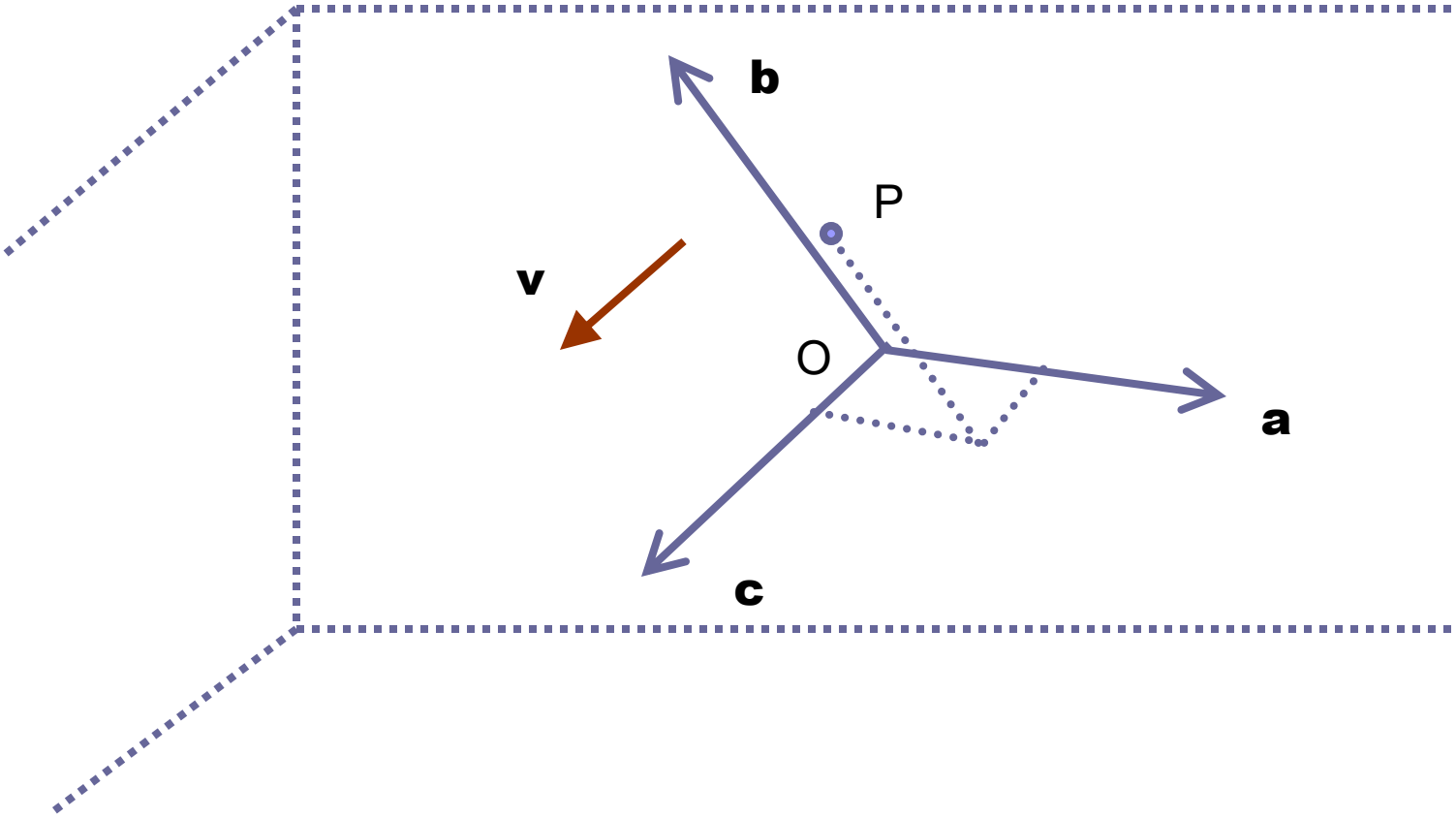
Application: Finding Normal to a Plane





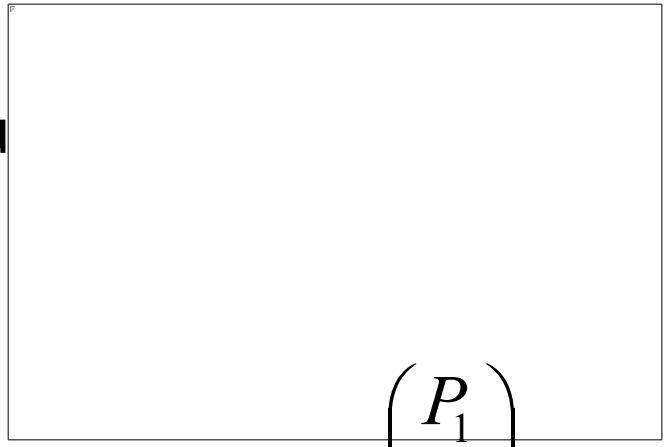
Representation of Vectors and Points

Coordinate Frame



Homogeneous Representation

- Basic objects: **a**,**b**,**c**,O
- Represent points and vectors u



$$\mathbf{v} = (\mathbf{a}, \mathbf{b}, \mathbf{c}, O)\begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ 0 \end{pmatrix}$$

$$P = (\mathbf{a}, \mathbf{b}, \mathbf{c}, O)\begin{pmatrix} P_1 \\ P_2 \\ P_3 \\ 1 \end{pmatrix}$$

Componentwise Linear Combinations

- Point – Point = Vector
- Point + Vector = Point
- Vector + Vector = Vector
- Scalar * Vector = Vector
- Scalar * Vector + Scalar * Vector = Vector

Point + Point = ??



Affine Combinations of Points

Linear combination of two points $P = (P_1,P_2,P_3,1)$ and $R = (R_1,R_2,R_3,1)$:

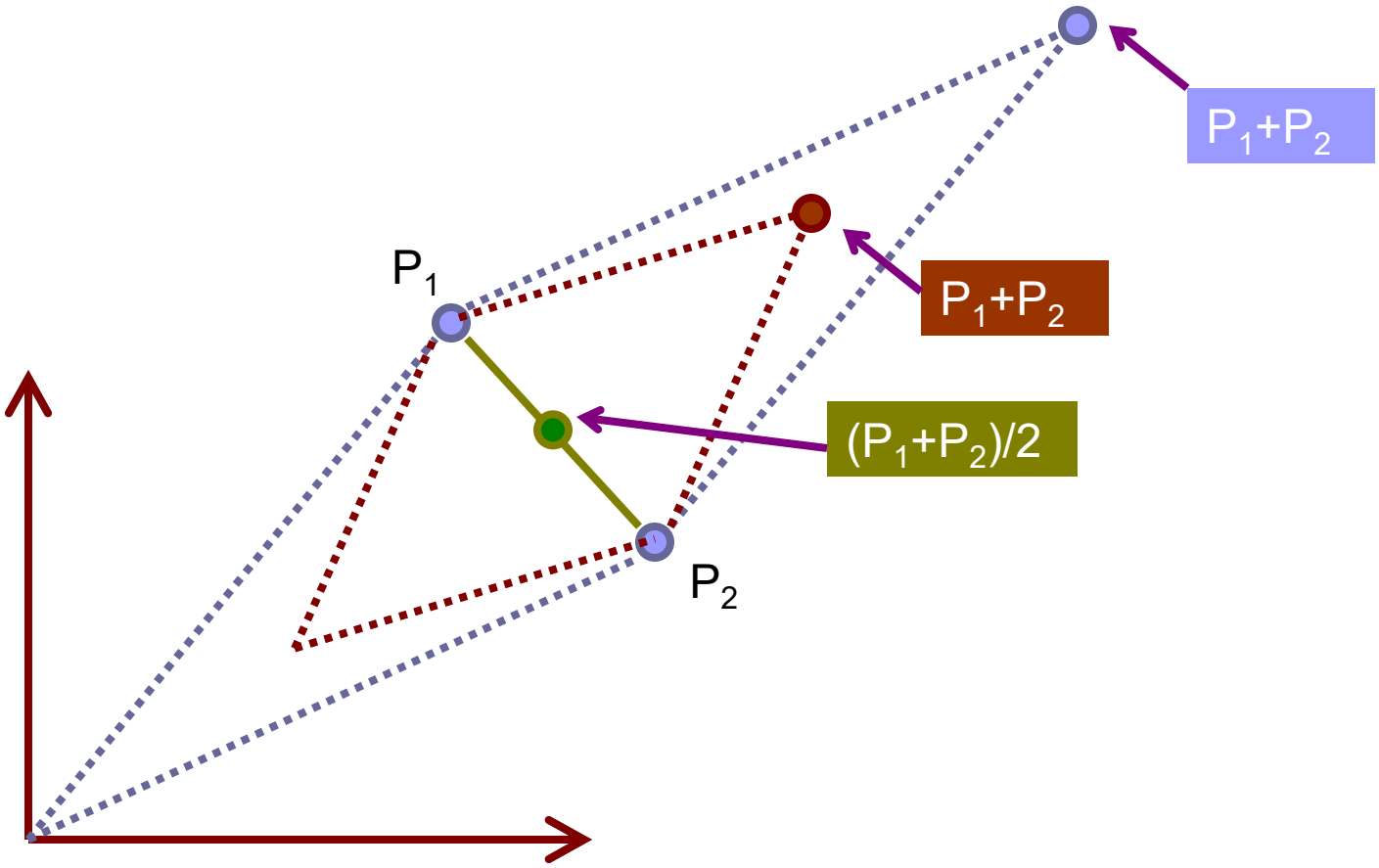
$$f\,P + g\,R = (f\,P_1 + g\,R_1,\ f\,P_2 + g\,R_2,\ f\,P_3 + g\,R_3,\ f + g)$$

NOT a valid point, unless $f + g = 1$

Example: 0.3P+0.7R is a valid point, but P + R is not.

any **affine combination** of points is a legitimate point

Affine Combinations of Points [contd]



Point + Vector = Affine Comb of Points

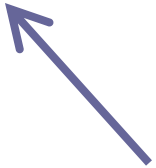
Let,

$P = A = t\,\mathbf{v}$ and $\mathbf{v} = B - A$

Therefore,

$P = A + t\,(B - A)$

$\Rightarrow P = t\,B + (1 - t)\,A.$

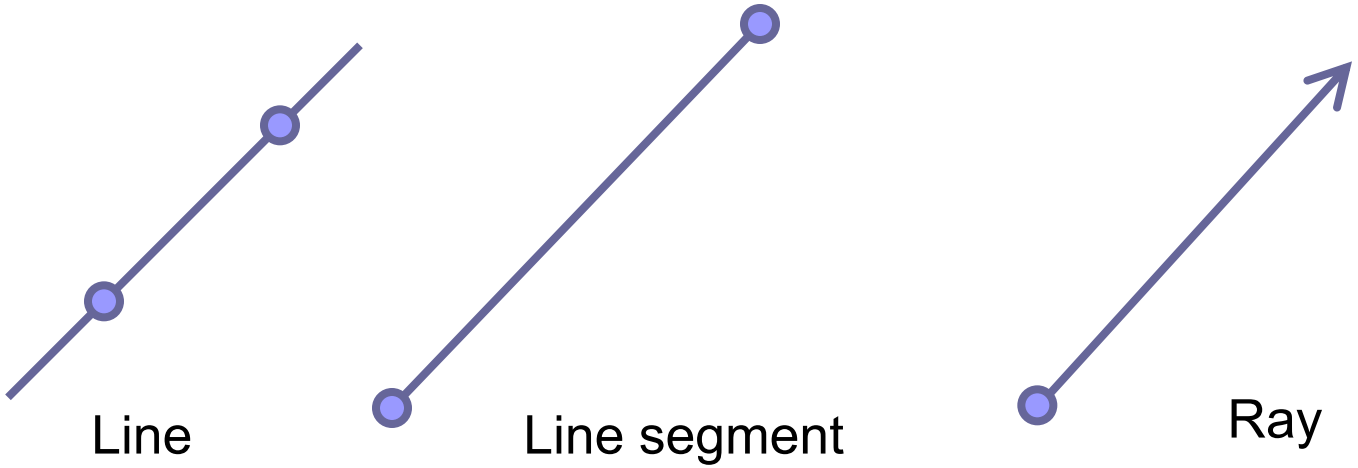


Linear Interpolation

"Tweening"

See Hill 4.5.4

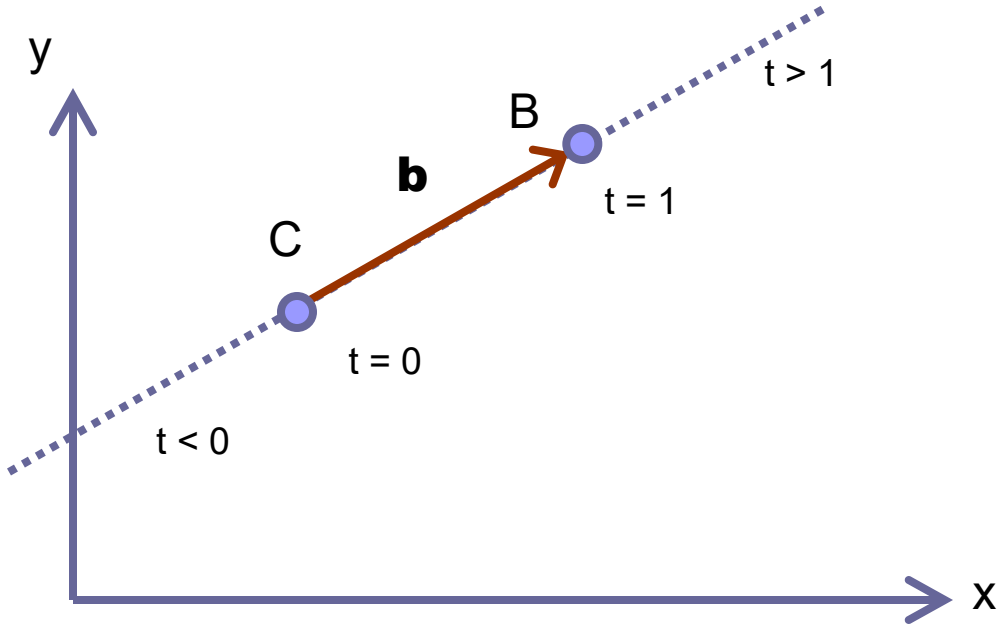
Representing Lines



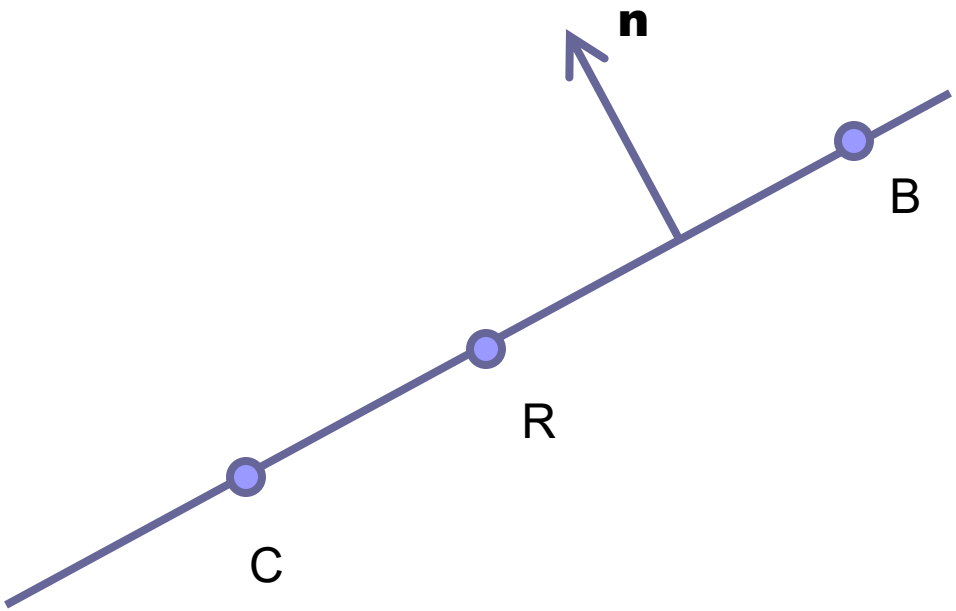
3 types of representations:

1. Two point form
2. Parametric representation
3. Point normal form

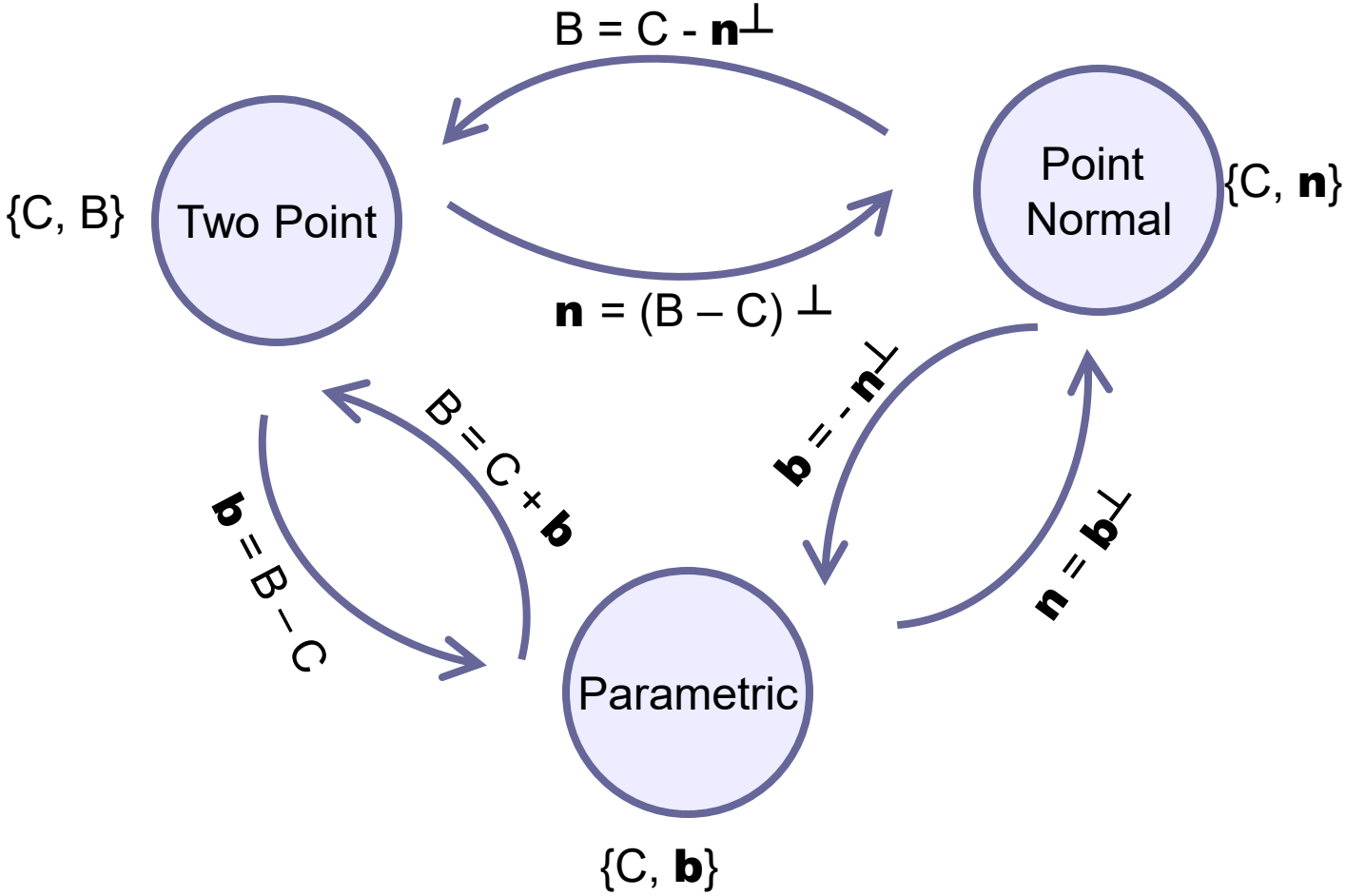
Parametric Representation of a Line



Point Normal Form of a Line



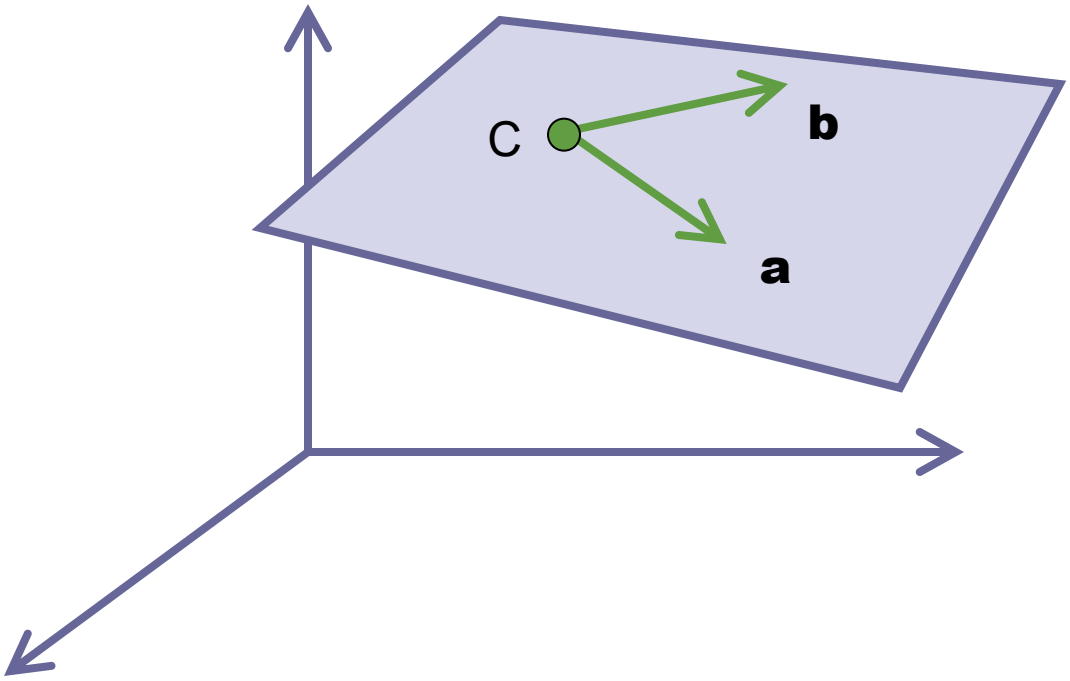
All in One : Representations of Line



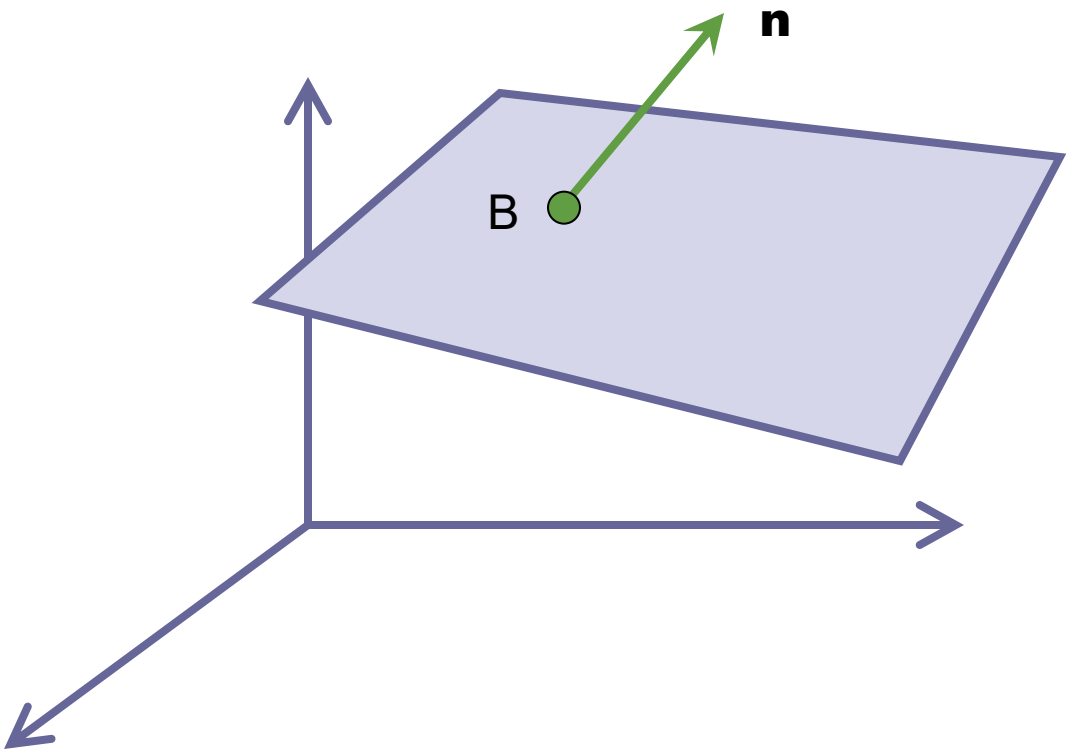
Planes in 3D

- 3 fundamental forms
  - Three-point form
  - Parametric representation
  - Point normal form

Parametric Representation of Plane



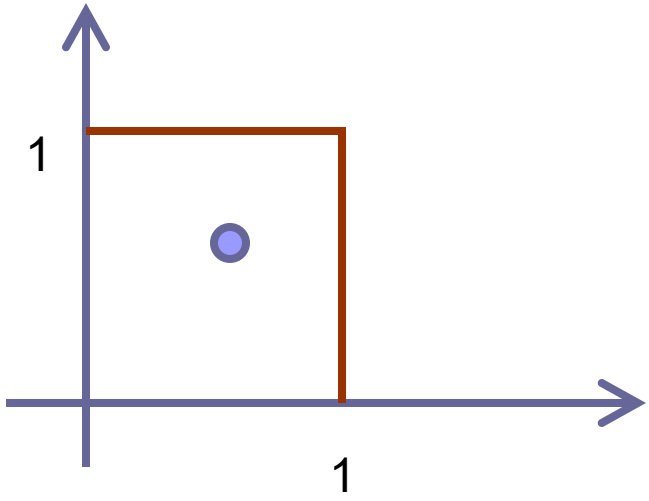
Point Normal Form of a Plane



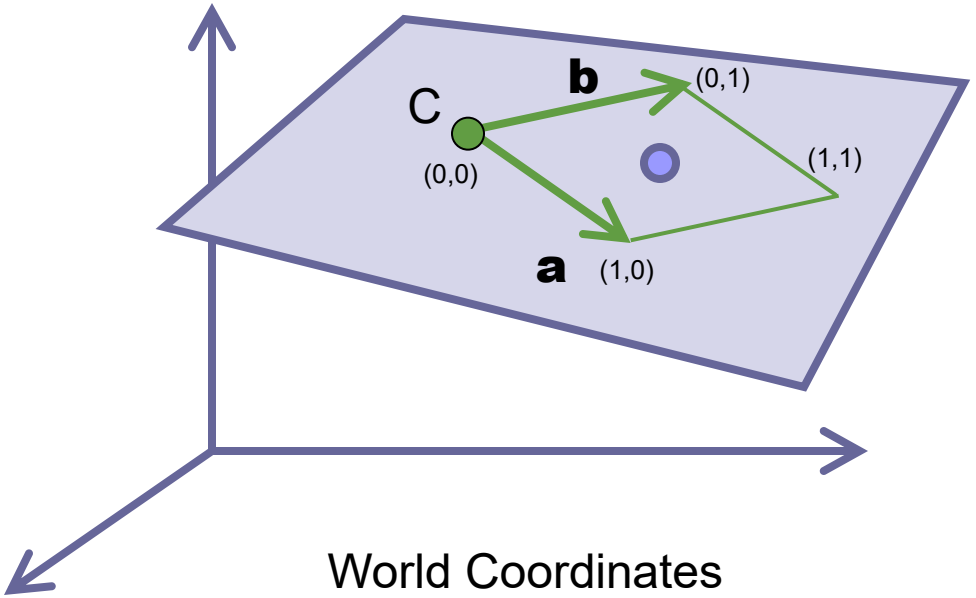
All in One : Representations of a Plane

See the Text Book!

Planar Patches



Parameter Space



World Coordinates